# DSA ki Baithak

CODE BAITHAK

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# Teaser / Trailer :

```
*   *   *
  *   *
*   *
  *
*
```

```
* * * * *
  * * * *
    * * *
      * *
        *
      * *
    * * *
  * * * *
* * * * *
```

```
    *
  *   *
*       *
*       *
  *   *
    *
```

```
4           4
4 3       3 4
4 3 2     2 3 4
4 3 2 1   1 2 3 4
4 3 2 1 0 1 2 3 4
4 3 2 1   1 2 3 4
4 3 2     2 3 4
4 3       3 4
4           4
```

```
        1
      2 1   1 2
    3 2 1     1 2 3
4 3 2 1         1 2 3 4
    3 2 1     1 2 3
      2 1   1 2
        1
```

```
5 5 5 5 5 5 5 5 5
5 4 4 4 4 4 4 4 5
5 4 3 3 3 3 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 2 1 2 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 3 3 3 3 4 5
5 4 4 4 4 4 4 4 5
5 5 5 5 5 5 5 5 5
```

# "Patterns Ka Gyaan...."

* Patterns always get printed rowwise (from left to right)
* Outer loop runs for the no. of rows

  &

  Inner loop runs for the no. of columns

# Patterns Halwa Recipe

Steps -

1. First find out the total no. of rows in the pattern.
2. Find the tasks to be performed in each row of pattern.
3. Perform all the tasks of the current row.
4. Do preparation of the next row & change the line.

11.

$n = 5$

| | | | | |
|---|---|---|---|---|
| - | - | - | - | * |
| - | - | - | * | * |
| - | - | * | * | * |
| - | * | * | * | * |
| * | * | * | * | * |

$n = 5$

① Rows $\rightarrow$ $n$

② Space, Star $\rightarrow$ row

| Space | Star |
|---|---|
| 4 | 1 |
| 3 | 2 |
| 2 | 3 |
| . | . |

③ ✓

④ ✓

12.

$n = 5$

```
a b c d e
_ a b c d
_ _ a b c
_ _ _ a b
        a
```

$n = 5$

① Rows → n

② space, chars → row

| space | chars |
|---|---|
| 0 | 5 |
| 1 | 4 |
| 2 | 3 |
| ⋮ | ⋮ |

③ ✓

④ ✓

13.

$n = 5$

| | | | | |
|---|---|---|---|---|
| - | - | - | - | 5 |
| - | - | - | 4 | 5 |
| - | - | 3 | 4 | 5 |
| | 2 | 3 | 4 | 5 |
| 1 | 2 | 3 | 4 | 5 |

① Rows → n

② space, Digit → row

| space | Digit |
|---|---|
| 4 | 1 |
| 3 | 2 |
| 2 | 3 |
| ⋮ | ⋮ |

③ ✓

④ ✓

14.

$n = 5$

```
* * * * *
_ _ * * * *
_ _ _ _ * * *
_ _ _ _ _ _ * *
            _ _ *
```

① Rows → n

② Space , Star → row

| Space | Star |
|---|---|
| 0 | 5 |
| 2 | 4 |
| 4 | 3 |
| 6 | 2 |
| 8 | 1 |

+2 ✓ -1 ✓

③

④

15.

$n = 5$

```
- - - - *
- - - * * *
- - * * * * *
  * * * * * * *
* * * * * * * * *
```

① Rows → n

② space, star → row

| space | star |
|---|---|
| 4 | 1 |
| 3 | 3 |
| 2 | 5 |

-1 +2

③ ✓

④ ✓

# Vertical Mirroring

16.

$n = 5$

```
        1
      2 1 2
    3 2 1 2 3
  4 3 2 1 2 3 4
5 4 3 2 1 2 3 4 5
```

digits/2

| digits | col |
|---|---|
| 1 | |
| 3 | |
| 5 | |
| 7 | |
| 9 | |

row - 4

4 3 2 1 2 3 4

val = 4
3
2
1

col = 1 2 3 4 5 6 7

4 3 2 1 2 3 4

7/2 = 3

col <= 3

# Horizontal Mirroring

17.

$n = 5$

```
- - - - *
- - - * * *
- - * * * * * -
- * * * * * * * -
* * * * * * * * *
  * * * * * * *
    * * * * *
      * * *
        *
```

① Rows → $2*n - 1$

② Space, Star → Row

| Space | Star |
|---|---|
| 4 | 1 |
| 3 | 3 |
| 2 | 5 |
| 1 | 7 |
| 0 | |

(-1) (+2)

| | |
|---|---|
| 1 | |
| 2 | |
| 3 | |

(+1) (-2)

row = 1 2 3 4 5

Star = 3 4 5 7 9 11

(row < = n)

4 <= 5

5 <= 5 ✓

18.

$n = 5$

```
        *
      * - *
    * - * - *
  *   *   *   *
*   *   *   *   *
  *   *   *   *
    *   *   *
      *   *
        *
```

row → 3

star → 5

even → -

odd → *

col = 1 *

col = 2 -

col = 3 *

col = 4 -

19.

$n = 5$

```
* * * * *
    * * * *
        * * *
            * *
                *
            * *
        * * *
    * * * *
* * * * *
```

20.

$n = 5$

```
        1
      1 2 1
    1 2 3 2 1
  1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
  1 2 3 4 3 2 1
    1 2 3 2 1
      1 2 1
        1
```

# Home Work - Question